



Robotics | Software Engineer | Mechatronics Engineer

Experience

December 2023 - July 2024

Data Science & New Technologies Consultant

[Management Solutions](#)

Digital transformation — retail banking automation.

I worked on a project to automate the process of credit risk assessment for a retail bank. I developed a machine learning model that predicts the probability of default for a customer based on their financial history and demographic information. I also worked on a project to develop a chatbot that can answer customer queries about their account and transactions. I used natural language processing techniques to train the chatbot to understand and respond to customer questions.

July 2023 - December 2023

Information and Technology Intern [World Food Programme | United Nations](#)
[Peru](#)

I worked as part of a small perception team developing the first ever autonomous delivery bot. My work focused on LiDAR scene segmentation, sensor fusion, and camera calibration. I was fully responsible for all software for cameras and LiDAR calibration.

January 2023 - June 2023

Application Engineer

[Andes Technology](#)

I was the first student at the lab and have set up all the lab infrastructure for a full CI setup. Scientifically I have worked on various topics such as motion planning, navigation, range-based scene interpretation, and mapping. I published at international conferences such as ICRA and IROS and my code is available as open source.

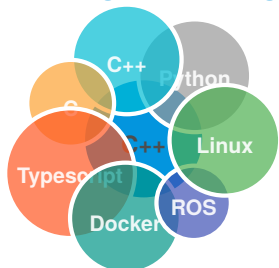
December 2021 - November 2022 **Industrial Engineering Assistant**

[Creditec](#)

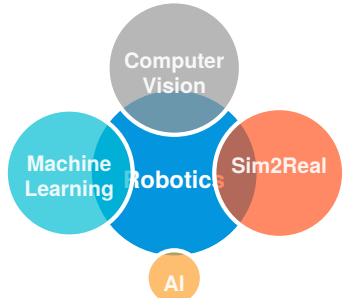
As a research assistant in the Autonomous Intelligent Systems lab at the University of Freiburg, led by Prof. Dr. Wolfram Burgard, I dealt with Kinect RGBD sensors and mapped them onto various platforms. I have implemented traversability analysis for a mobile robot as part of the ROVINA project. The developments in this project led to a publication already during my masters.

Selected Projects

Programming



Fields of Interest



Languages

English [bilingual] ★★★★★
French [proficient] ★★★★★
Spanish [native] ★★★★★

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2026 - Present | **Modern Robot Learning**

[AI Lab - 42 AI](#)

A modular robotics research platform for learning and experimentally evaluating modern approaches to robot perception, manipulation, imitation learning, reinforcement learning, and embodied foundation models.

<https://github.com/cuycorp/Modern-Robot-Learning>

2026 - Present | **Duck - a thought tracker**

[Personal project](#)

A local-first focus companion, desktop overlay application, that lets users quickly externalize distracting thoughts and return to their task.

<https://github.com/cuycorp/Duck>

2026 - Present | **Tokenscope**

[Ecole 42](#)

TokenScope is a collaborative platform that helps engineering teams understand where their LLM spending comes from by collecting usage traces, calculating estimated cost, visualizing live analytics, and generating grounded optimization recommendations.

<https://github.com/lynyam/tokenscope>

2025 | **Minishell**

[Ecole 42](#)

Developing an autonomous robot for underground exploration I was responsible for traversability analysis, navigation, exploration and some of the mapping tasks. The project received excellent reviews from the EU commission.

<https://github.com/cuycorp/minishell>

Honors & Awards

2013 **MINT Excellence Network Member**

[MLP Stiftung](#)

I was chosen as one of 300 best applicants across Germany to the MINT Excellence Network.

The candidates were chosen from the students who work in the fields of Maths, Computer Science, Natural Sciences and Tech across Germany.

2018 **Best PhD Thesis of the Faculty Award**

[University of Bonn](#)

My PhD thesis "Robot mapping and navigation in real-world environments" has been awarded the title of the best out of all written at my faculty in 2018.

Education

2011 - 2014 **MSc. Computer Software Engineering**

[Ecole 42, France](#)

A project-based and peer-to-peer computer science program ranked the world's 3rd most innovative university in the 2025 and 2026 WURI rankings. Training focused on system-level programming, C, Linux, Git, and shell development, with extensive team projects in an autonomous, collaborative learning environment.

2017- 2022 **BSc. Mechatronics Engineering**

[Pontifical Catholic University of Peru, Lima, Peru](#)

The Pontifical Catholic University of Peru (PUCP) is ranked No. 1 in Peru and No. 10 in Latin America. My studies focused on mechanical and electronic systems and the design of integrated hardware–software automation systems. Thesis: design of an EEG-controlled omnidirectional wheelchair..

2015 - 2016 **Bilingual International Baccalaureate Diploma**

[Newton College, Lima, Peru](#)

Top bilingual Peruvian-British school, focused on developing inquisitive, outspoken and well-rounded individuals that have intercultural understanding.

September 28, 2026

CAMARENA PUCUHUANCA, Monica Francesca